Abstract

1 Background

2 Methods

2.1 Sequential per-segment Newton controller

The scan path is divided into control blocks that are optimized in execution order. The control vector for block i is given in Equation (1).

$$\mathbf{u}_i = (P_{\mathrm{mod},i},\, \sigma_i,\, v_i)^{\mathsf{T}}, \qquad P_i = P_0 P_{\mathrm{mod},i}, \tag{1}$$

Here, $P_{\mathrm{mod},i}$ is the absorbed-power multiplier, σ_i is the lateral beam standard deviation, and v_i is the scan velocity. The melt-pool half-width w_i (the transverse semi-span from the scan centreline to the liquidus boundary) and depth d_i for a given set of control parameters $\mathbf{u}_i$ are evaluated at the end of the block and compared with the developed single-track targets (w^*, d^*) through the normalized residual given in Equation (2).

$$\widetilde{\mathbf{r}}_i(\mathbf{u}_i) = \begin{pmatrix} (w_i - w^*)/w^* \\ (d_i - d^*)/d^* \end{pmatrix}. \tag{2}$$

Figure 1 summarizes this causal block update: a change made at x_i is judged by how closely the melt-pool dimensions at x_{i+1} approach their targets.

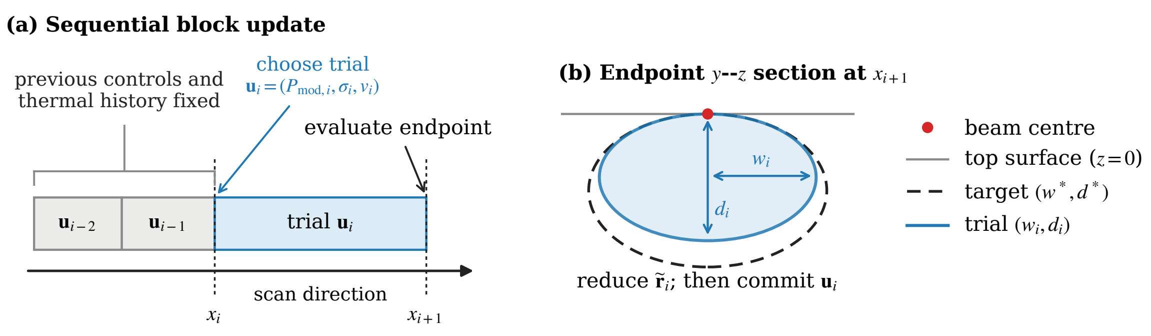

Figure 1: Sequential control-block update. Previously accepted controls define the fixed thermal history up to x_i. A trial control $\mathbf{u}_i$ is applied over $[x_i, x_{i+1}]$, and its endpoint width and depth are compared with the single-track targets in a transverse y–z melt-pool section.

The controls already accepted for blocks $j < i$ are replayed as fixed thermal history, so the update for block i cannot alter any preceding portion of the scan. The approach is therefore a greedy optimization: the controls at block i are chosen to give the best possible melt-pool dimensions at its endpoint x_{i+1}, then frozen before block $i+1$ is optimized. Since each block looks only one block ahead and never revisits an earlier decision, the schedule is optimal block-by-block but not necessarily globally.

The temperature field and its derivatives with respect to the current block's power, beam width, and velocity are evaluated in one simulation using first-order order-truncated imaginary (OTI) arithmetic. Spatial temperature gradients and control derivatives on the liquidus isotherm are converted to derivatives of w_i and d_i using the implicit function theorem. This gives the Jacobian in Equation (3).

$$J_i = \begin{pmatrix} \partial w_i/\partial P_{\mathrm{mod},i} & \partial w_i/\partial \sigma_i & \partial w_i/\partial v_i \\ \partial d_i/\partial P_{\mathrm{mod},i} & \partial d_i/\partial \sigma_i & \partial d_i/\partial v_i \end{pmatrix}. \tag{3}$$

Residuals and controls are scaled, with $S = \mathrm{diag}(1, 200\ \mu\mathrm{m}, 3\ \mathrm{m\,s^{-1}})$ collecting the nominal magnitude of each control (Table 1), so that the scaled step $\Delta\mathbf{z}$ is dimensionless and a single λ regularizes the three controls evenly. A Levenberg–Marquardt (regularized Gauss–Newton) step is then computed as in Equation (4).

$$(\widetilde{J}_i^{\mathsf{T}} \widetilde{J}_i + \lambda I)\Delta\mathbf{z} = -\widetilde{J}_i^{\mathsf{T}} \widetilde{\mathbf{r}}_i, \qquad \Delta\mathbf{u}_i = S\Delta\mathbf{z}, \tag{4}$$

where $\lambda = 2.5 \times 10^{-2}$ and $\widetilde{J}_i$ is the target-normalized Jacobian in scaled control coordinates, so that $\widetilde{J}_i\,\Delta\mathbf{z}$ is precisely the first-order change in the normalized residual $\widetilde{\mathbf{r}}_i$ produced by the scaled step $\Delta\mathbf{z}$. Since the update is underdetermined, the width–depth residual alone does not determine a unique step: a family of $(P_{\mathrm{mod}}, \sigma, v)$ corrections can produce the same first-order geometric change. Thus, a small penalty is placed on departures from the nominal velocity, which resolves this degeneracy by preferring the correction closest to nominal speed, without allowing a worse geometric residual to be accepted.

The update is safeguarded by componentwise per-iteration step-size caps of 0.5 in P_{mod}, 80 μm in σ, and 1.5 $\mathrm{m\,s^{-1}}$ in v, together with the bounds

$$0.05 \leq P_{\mathrm{mod}} \leq 3, \qquad 80 \leq \sigma \leq 400\ \mu\mathrm{m}, \qquad 0.5 \leq v \leq 8\ \mathrm{m\,s^{-1}}. \tag{5}$$

Because the step is derived from a linearized model, its full length can step beyond the region where the linearization is accurate, so the predicted reduction in the residual is not realized. A trial control $\mathbf{u}_i + \alpha\,\Delta\mathbf{u}_i$ is therefore formed with a backtracking step-length factor $\alpha \in \{1, 1/2, 1/4, 1/8, 1/16\}$, and the first value that decreases the physical merit $\frac{1}{2}\|\widetilde{\mathbf{r}}_i\|_2^2$ is accepted; if none does, no update is taken. A block is accepted without an OTI evaluation if a preliminary double-precision replay is already within a 5% endpoint tolerance; otherwise, at most six Newton iterations are taken.

The first block after each turnaround is initialized with a deliberately cool, slow guess ($P =$ 30 W, $\sigma = 200\ \mu$m, and $v = 1\ \mathrm{m\,s^{-1}}$). Because the material just past a turnaround is still hot from the adjacent track, warm-starting this block from the preceding high-speed controls would pin the Newton iteration in a spurious high-velocity local solution; the cool, slow guess avoids that basin. Each later block on the same track is instead warm-started from the immediately preceding optimized block, promoting smooth control transitions along the scan. The accepted controls are then committed to the scan history before the next block is optimized.

3 Results

The studies below use the common material and nominal process conditions in Table 1. These conditions match those used by Stump and Plotkowski [1]. Parameters are varied only where stated. Melt-pool widths are reported here as the full transverse span, i.e. twice the half-width w_i regulated by the controller; the normalized residual and Jacobian make this a matter of convention only.

parameter	symbol	value
Material and thermal properties		
thermal conductivity	k	26.6 $\mathrm{W\,m^{-1}K^{-1}}$
specific heat capacity	c_p	600 $\mathrm{J\,kg^{-1}K^{-1}}$
density	ρ	7451 $\mathrm{kg\,m^{-3}}$
initial temperature	T_0	1273 K
liquidus temperature	T_{liq}	1610 K
Nominal process conditions		
absorbed beam power	P_0	150 W
lateral beam standard deviation	$\sigma_{xy,0}$	200 μm
absorption-depth standard deviation	σ_z	10 μm
scan speed	v_0	3 $\mathrm{m\,s^{-1}}$
hatch spacing	h	0.1 mm
tracking timestep	Δt	50 μs

Table 1: Material properties and nominal process conditions used throughout the two-track, square, and triangle studies.

3.1 Two-track base case

The two-track case is the smallest geometry considered that contains both a single turnaround and thermal interaction between neighbouring tracks. It therefore provides a compact base case for evaluating the optimization procedure, isolating turnaround behaviour, and selecting an appropriate control-block length before moving to full raster geometries. The simplest control policy is considered: continuous serpentine scanning in which only the power P, lateral beam width σ, and scan velocity v are modulated, compared against the unoptimized nominal serpentine. Each scan line is divided into uniform control blocks, and the block length is varied from 5 to 0.1 mm. This sweep identifies 1 mm as a suitable working block length for the larger square and triangle studies, as demonstrated in Table 2.

control block	tracked depth (μm)
unoptimized baseline	69.4 ± 12.4
5 mm	65.1 ± 9.4
2.5 mm	63.4 ± 7.8
1 mm	$\mathbf{62.5 \pm 6.6}$
0.5 mm	68.4 ± 10.3
0.1 mm	101.6 ± 16.3

Table 2: Two-track block-length sweep for the continuous $P/\sigma/v$ policy: beam-on tracked melt-pool depth (mean $\pm$ standard deviation) about the developed single-track target. Regulation is tightest at 1 mm; the tested 0.5 and 0.1 mm schedules destabilize.

The spatial outcome, optimized control history, and transient response of the selected 1 mm controller are shown in Figures 2–4. Relative to the nominal serpentine, the tracked depth standard deviation falls from 12.4 to 6.6 μm and the surface fusion-depth standard deviation from 9.2 to 4.4 μm.

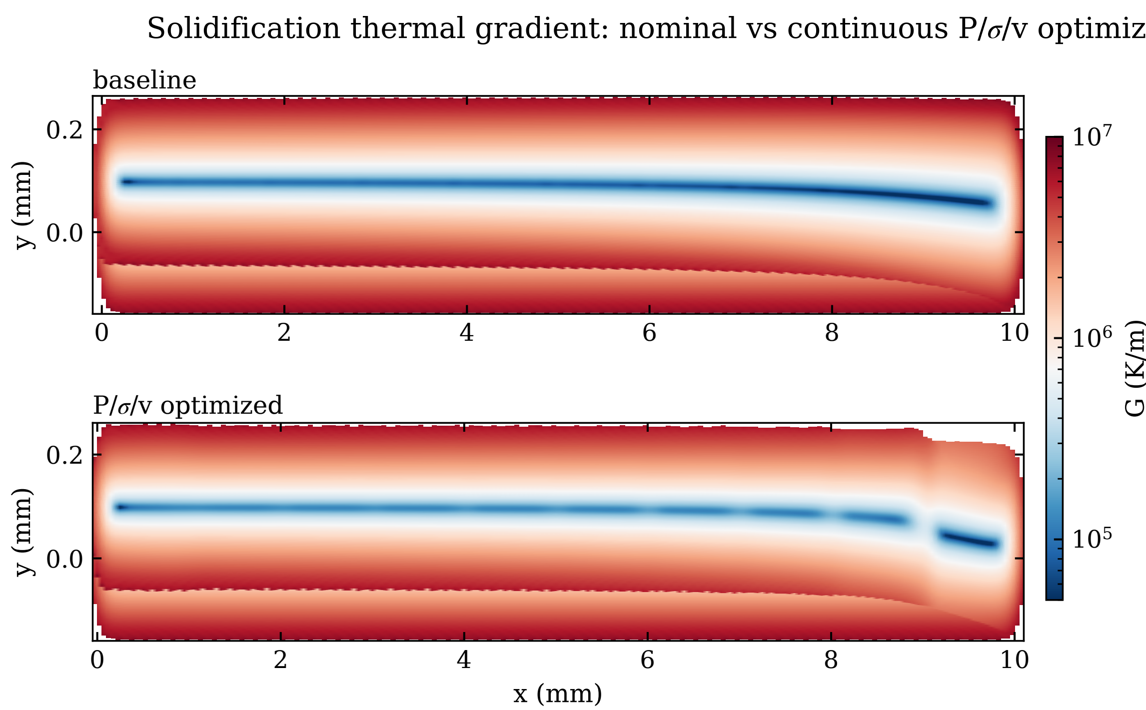

Figure 2: Surface solidification thermal-gradient maps for the nominal serpentine (top) and the 1 mm continuous $P/\sigma/v$ optimized schedule (bottom), on a common logarithmic colour scale.

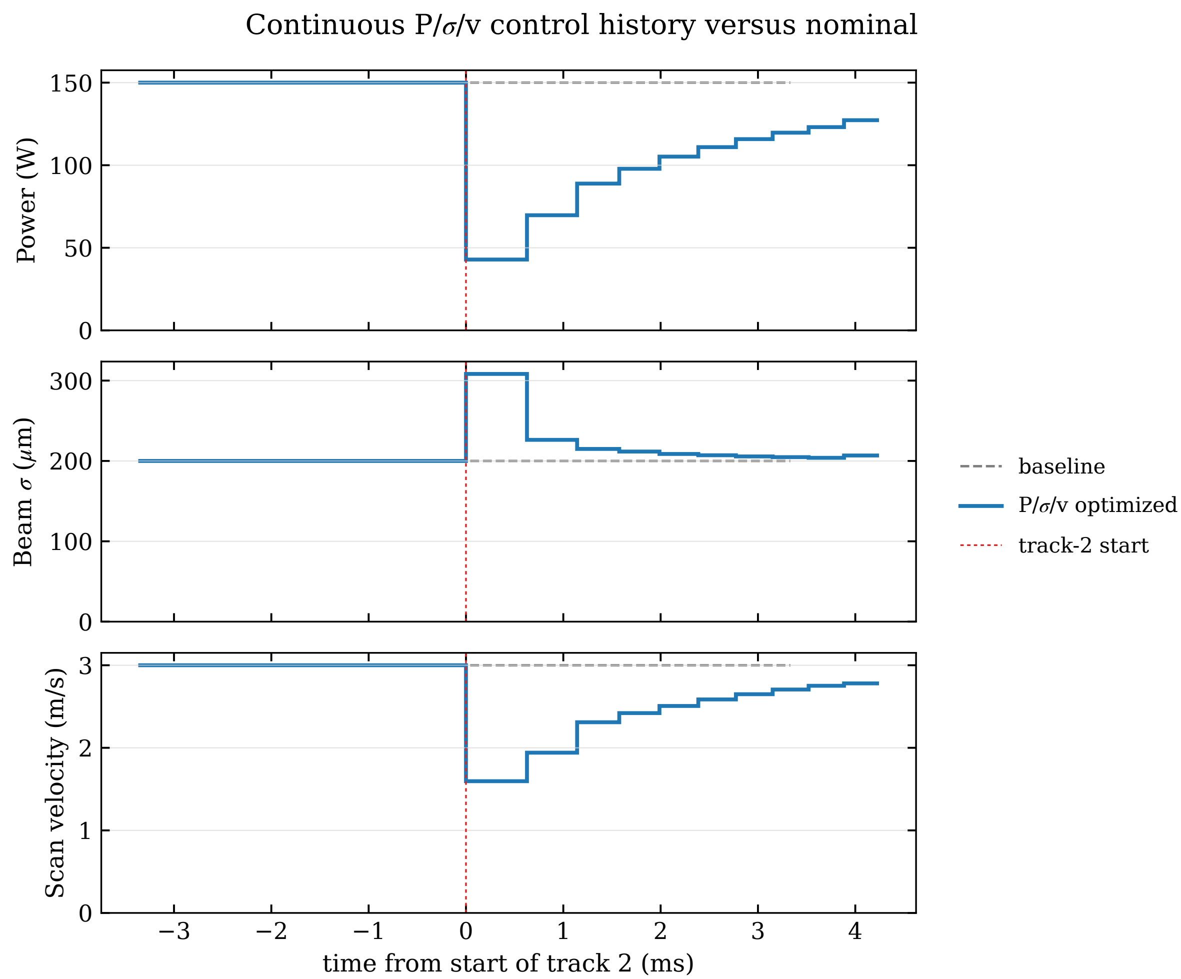

Figure 3: Power, lateral beam width, and scan-speed histories for the 1 mm continuous schedule against the nominal baseline. The track-2 start is aligned at zero; the controller slows and cools at the turnaround, then ramps smoothly back toward nominal.

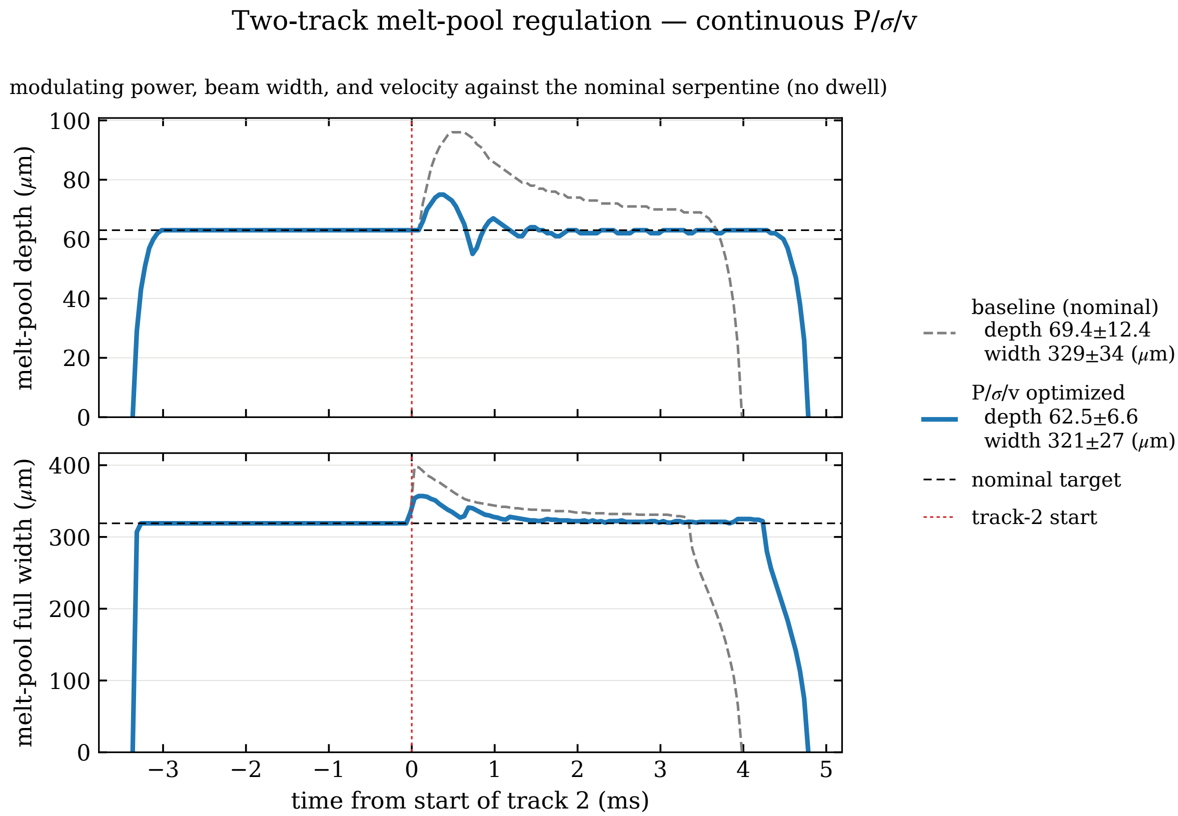

Figure 4: Melt-pool depth and full-width traces for the nominal baseline and the selected 1 mm continuous $P/\sigma/v$ controller. The baseline overshoots at the turnaround and decays slowly; the optimized schedule holds the developed target. The complete block-length sweep is reported in Table 2.

3.2 Square raster

The 10×10 mm square is the first full raster geometry considered and consists of 101 equal-length scan tracks. Continuous serpentine scanning is used throughout, with the power, lateral beam width, and scan velocity modulated over 1 mm control blocks. This case tests whether the melt-pool dimensions can be controlled in a simple raster geometry.

schedule	depth (μm)	full width (μm)	volume (mm^3)
baseline	106.0 ± 12.7	388 ± 27	0.0798 ± 0.0240
optimized	$\mathbf{62.2 \pm 2.5}$	$\mathbf{323 \pm 9.9}$	$\mathbf{0.01293 \pm 0.00127}$

Table 3: Square-raster beam-on melt-pool statistics under continuous serpentine scanning (mean $\pm$ standard deviation) from full tracked replays on the 50/10/1 μm grid.

The optimized schedule brings the mean depth and width to the single-track targets while reducing their variation. The depth standard deviation falls from 12.7 to 2.5 μm and the full-width standard deviation falls from 27 to 9.9 μm.

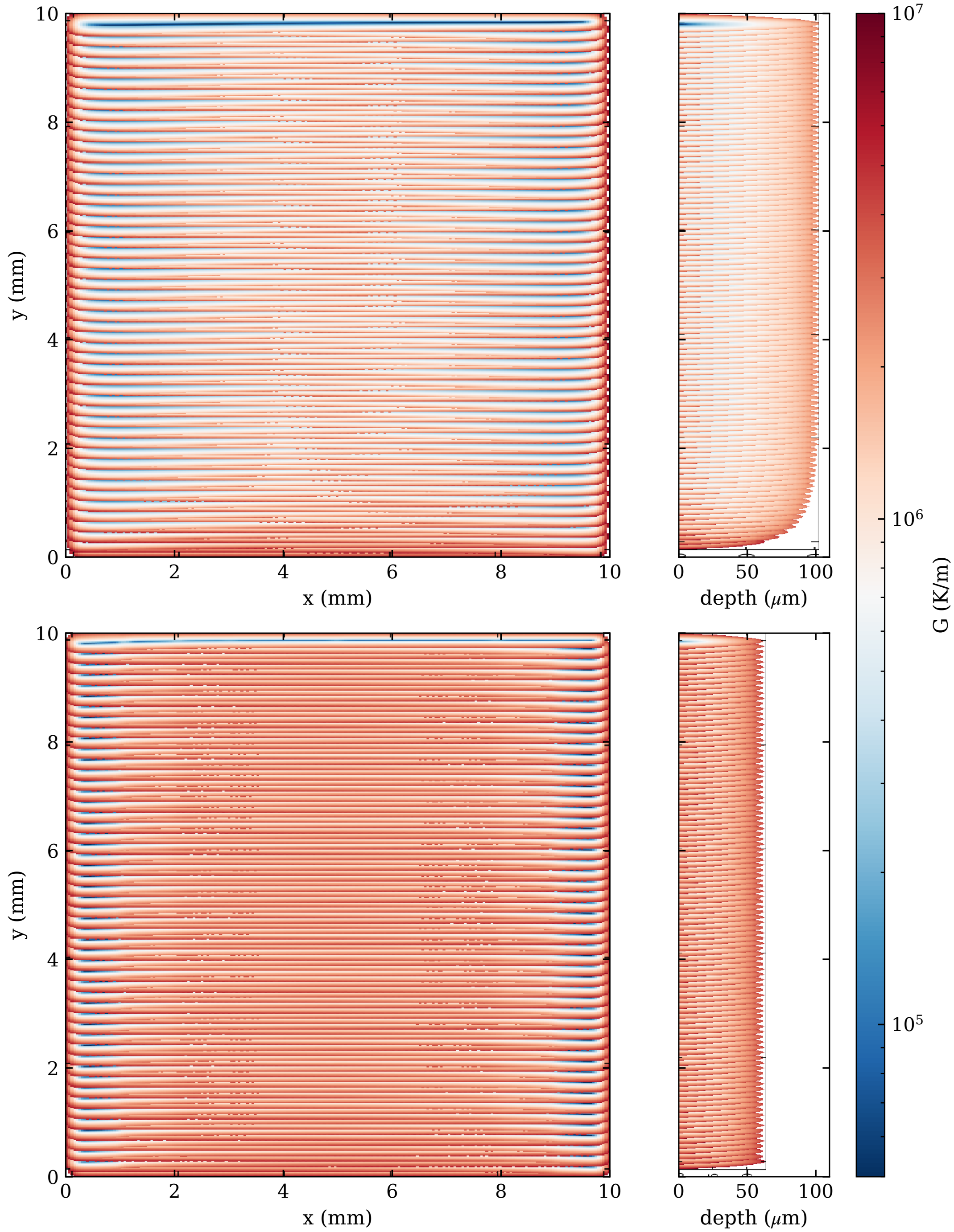

Figure 5: Square-raster solidification thermal-gradient maps under continuous serpentine scanning. The top-surface map (left) and vertical centre section (right) are shown for the baseline (top) and 1 mm optimized (bottom) schedules.

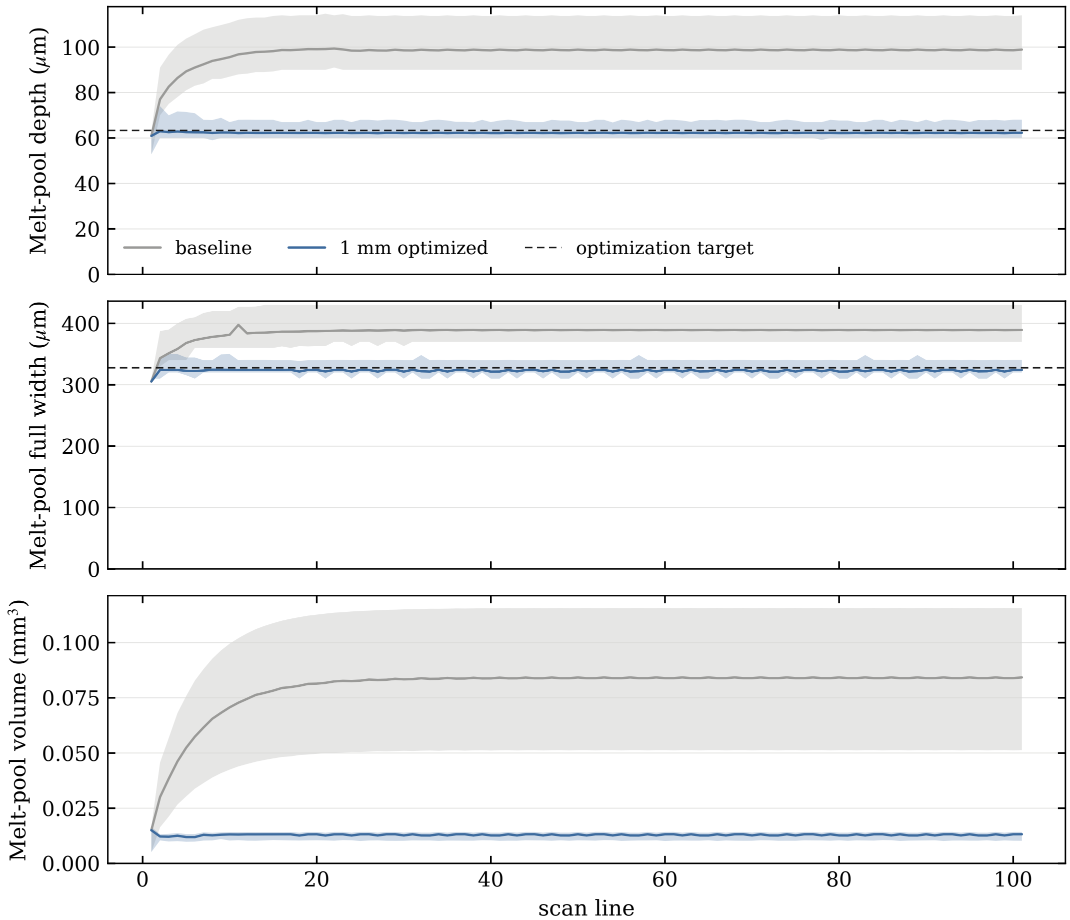

Figure 6: Square-raster per-line melt-pool statistics under continuous serpentine scanning for the baseline and 1 mm optimized schedules.

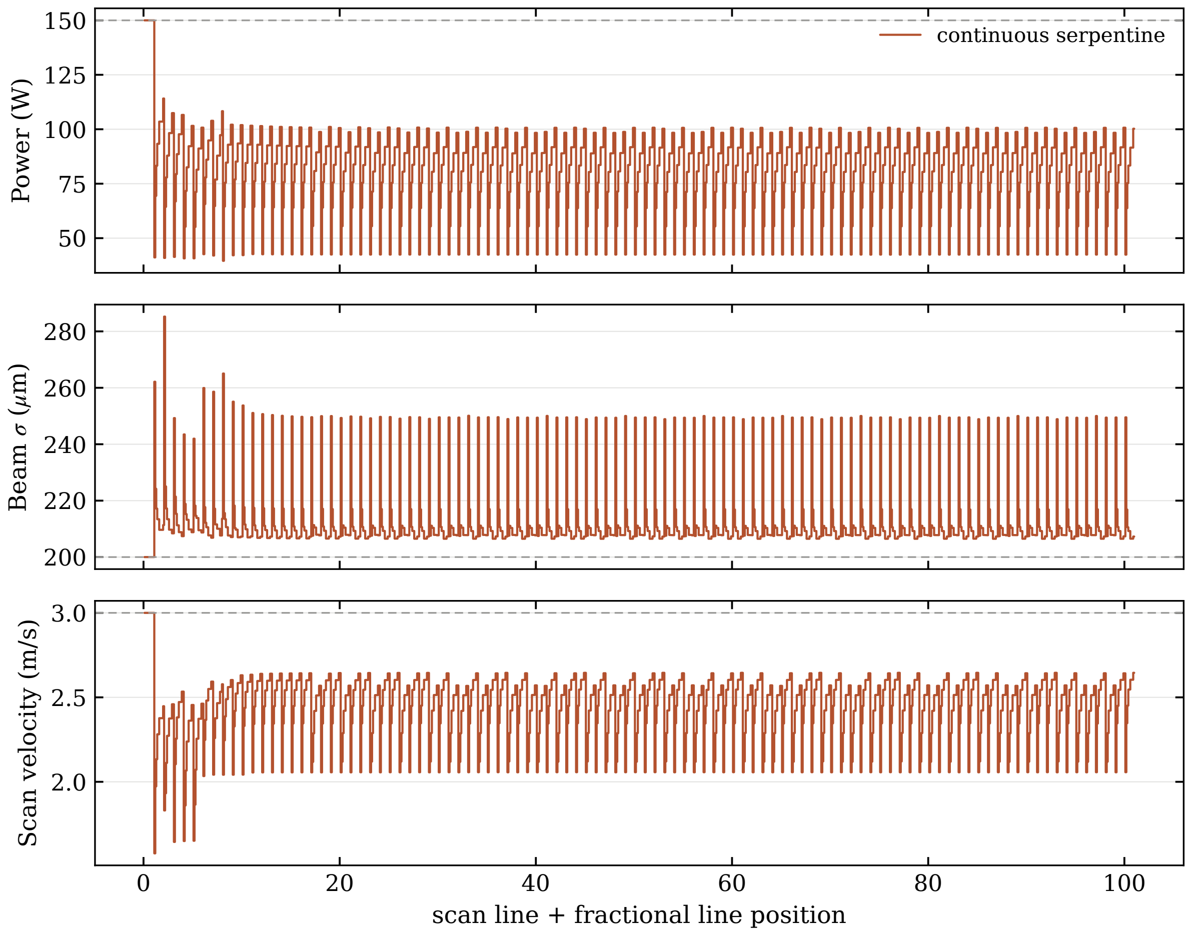

Figure 7: Optimized power, lateral beam width, and scan-speed histories for the square raster under continuous serpentine scanning.

3.3 Triangle raster

The equilateral triangle has 10 mm sides and consists of 87 scan tracks whose lengths decrease towards the apex. Continuous serpentine scanning is used throughout, with the power, lateral beam width, and scan velocity modulated over nominally 1 mm control blocks. This case tests whether the melt-pool dimensions can be controlled as the scan-track length progressively decreases. The blocks on each track are balanced so that a short remainder is not left at the end of the track.

schedule	depth (μm)	full width (μm)	volume (mm^3)
baseline	118.7 ± 21.2	484 ± 170	0.11274 ± 0.04715
optimized	$\mathbf{63.7 \pm 4.4}$	$\mathbf{329 \pm 15}$	$\mathbf{0.01176 \pm 0.00160}$

Table 4: Triangle-raster beam-on melt-pool statistics under continuous serpentine scanning (mean $\pm$ standard deviation) from full tracked replays on the 50/10/1 μm grid.

The optimized schedule brings the mean depth and width to the single-track targets while reducing their variation. The depth standard deviation falls from 21.2 to 4.4 μm and the full-width standard deviation falls from 170 to 15 μm.

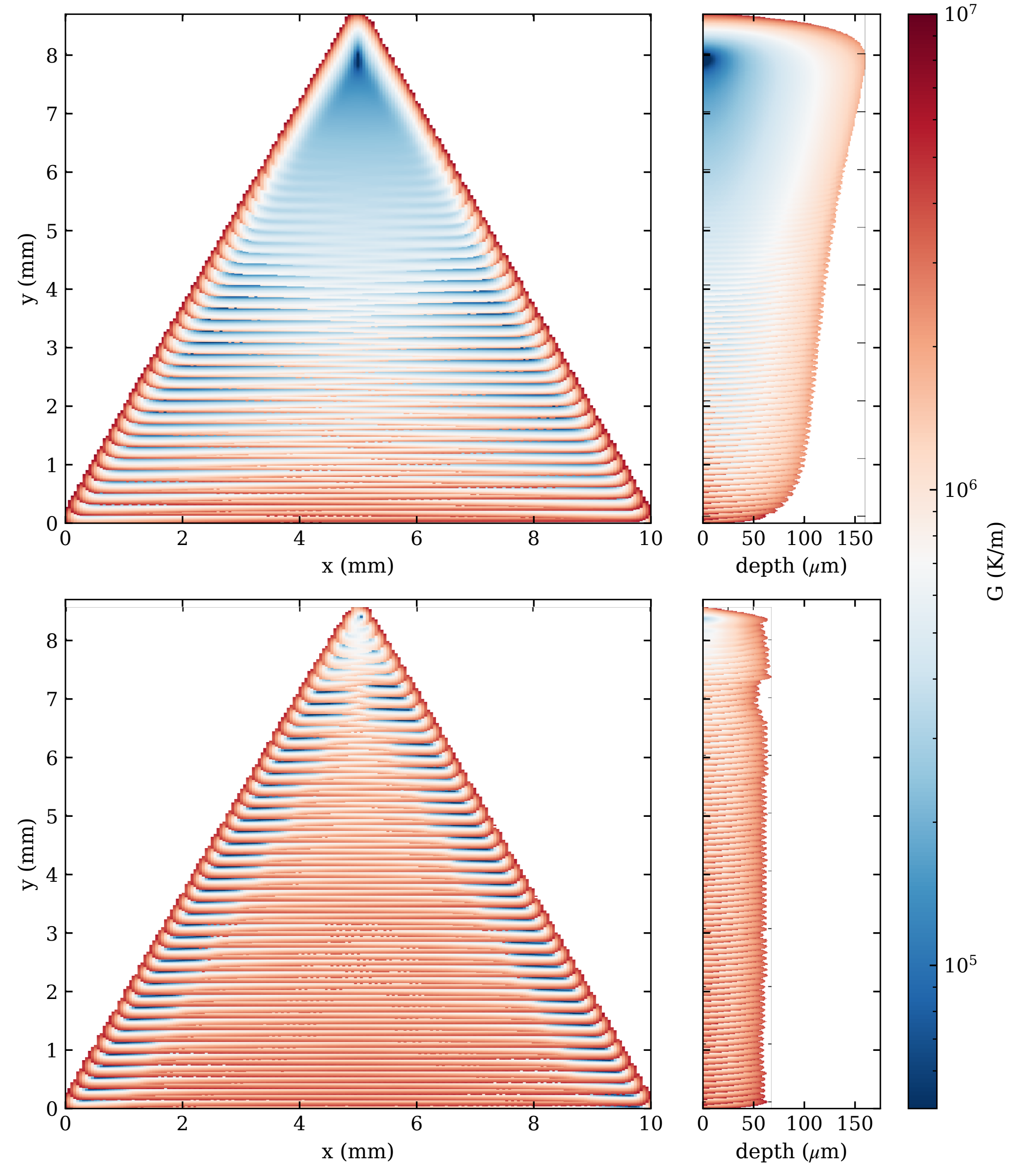

Figure 8: Triangle-raster solidification thermal-gradient maps under continuous serpentine scanning. The top-surface map (left) and vertical centre section (right) are shown for the baseline (top) and 1 mm optimized (bottom) schedules.

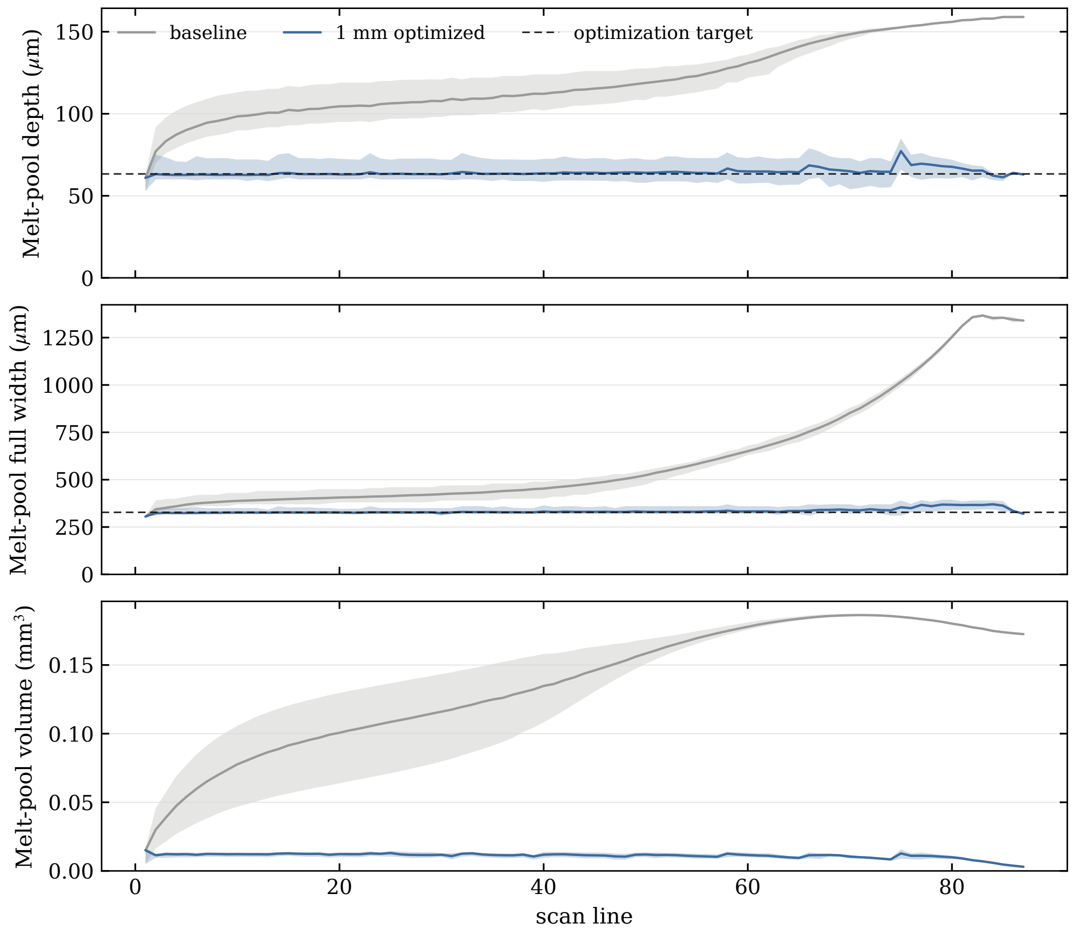

Figure 9: Triangle-raster per-line melt-pool statistics under continuous serpentine scanning for the baseline and 1 mm optimized schedules.

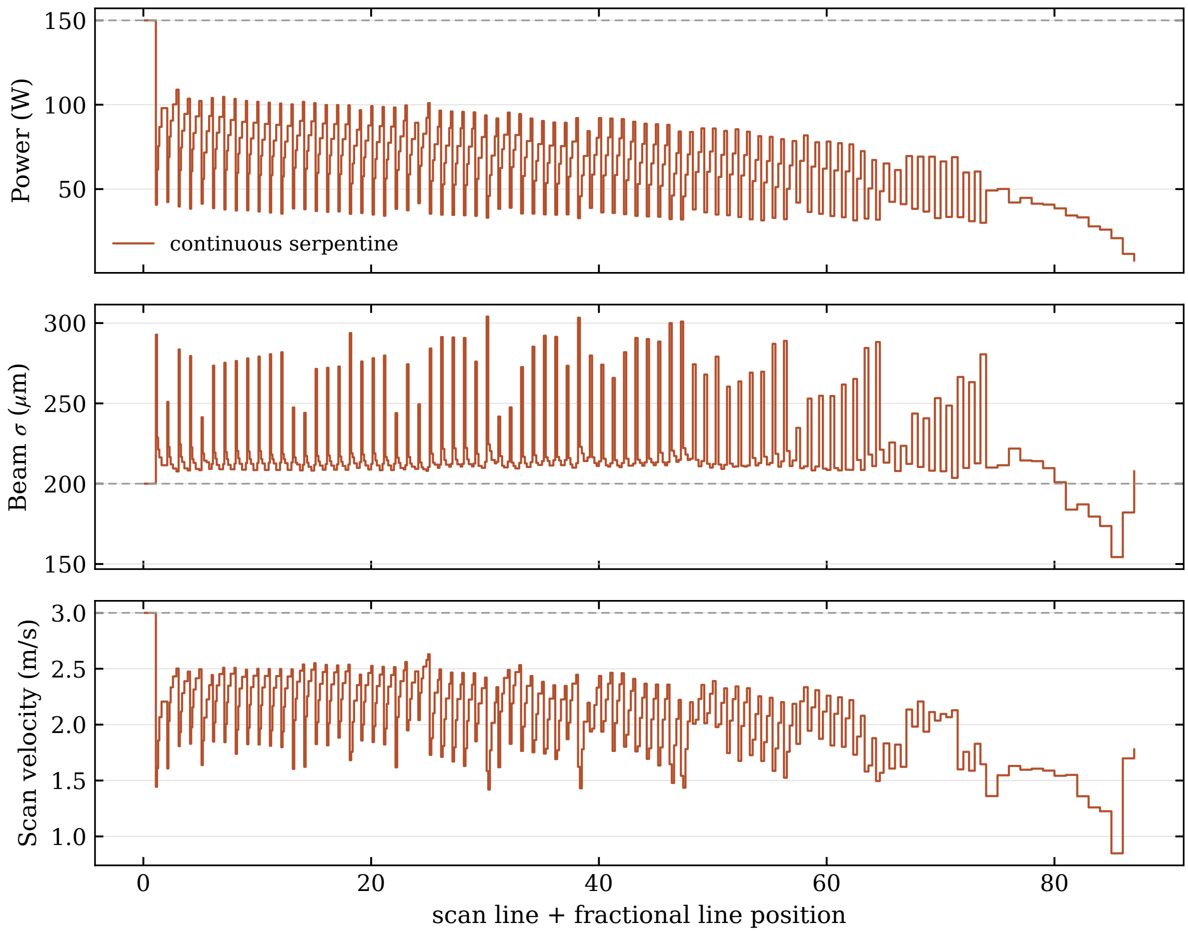

Figure 10: Optimized power, lateral beam width, and scan-speed histories for the triangle raster under continuous serpentine scanning.

4 Discussion

The studies above use continuous serpentine scanning and modulate only the continuous process controls—power, beam width, and scan velocity. A natural extension is to admit the turnaround dwell as an additional free control, so that the optimizer sizes a beam-off interval alongside the continuous parameters. This more elaborate strategy is left to a future study.

5 Conclusion

References

[1] B. Stump and A. Plotkowski. An adaptive integration scheme for heat conduction in additive manufacturing. *Applied Mathematical Modelling*, 75:787–805, 2019.